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Circular

Toxicological emergencies are the exigencies (an urgent need) which are encountered invariably by all Veterinary institutes rendering the health cover & Husbandry services. The prime institutes like Regional Disease Investigation Laboratories (7), District Veterinary polyclinics (29) /Taluka Veterinary Mini-Polyclinics (172) and Veterinary dispensaries, aid centers under local sector are expected to render specialized aid on spot in poisoning cases.

It is felt necessary to update the Circular No - DIS/Toxi/TE-Kit/7044-7160/1992, dated - 24/07/1992 because of the fact that, since last 10 -15 years, the movement of biological/Livestock, use of fertilizers, insecticides/pesticides has increased. This may be a one of the major reason for comparable increase in the cases of poisoning in the field. Whatever may be the reasons or the predisposing factors responsible for the poisoning but the fact can not be denied.

Therefore it was under consideration of this Laboratory to revise this circular, so that it can brush up the knowledge of fresh/experienced field Veterinarians, which will be of immense help in combating the cases of poisoning. It is therefore desirable that, the Veterinary institutes should have these kits ready, fully replenished from time to time and updating the contents after use.

The enclosed appendix is only a guideline as to what essential medicaments and equipments are required to prepare a "Toxicological emergency kit". Uses of Kit depend upon the local requirements and there may be change in need of these Kit. Most of the medicaments as per the line of treatment of a particular poisoning are already in use by Veterinary institutes.

Keep up such kit fully equipped and in portable readiness would enhance the efficacy and improve the utility of the services rendered at such Veterinary institutes. Besides, such timely and effective help to save the poisoned animals, would be appreciable by the livestock owners.

Treatment of Ailing Cases of Common Poisoning in Livestock

Animal toxicities/poisonings are a frequent presenting complaint in veterinary emergency clinics and general practice. Acute poisonings represent a diagnostic and therapeutic challenge for veterinary practitioners, whether single or multiple animals are involved. *The key to success in treating affected animals with toxicity/poisoning is early detection and treatment.*

Prognosis varies considerably, depending on the toxin involved, the amount of exposure, and the length of time that has elapsed between the exposure and presentation to the veterinarian. The general approach to the poisoned animal includes stabilization of vital signs and institution of supportive care measures, **obtaining a detailed clinical history** and samples of the suspected toxic substance or source (e.g. feed samples, plants etc.) if possible, decontamination to prevent further absorption of the toxicant, enhancing elimination of absorbed toxicants and administering a specific antidote.

1. Just/merely sending the samples makes it inconvenient for the Toxicological laboratory investigations, so also it does not serve any purpose.
2. **In general Required History -**
 - a. Your letter shall indicate required history.
 - b. You are aware of the fact that the 'Poisoning' is a broad term. There are number of poisons; may be of plant/shrubs/grass origin, organic/inorganic chemical origin, moulds, Aflatoxin etc.
 - c. Therefore, for Toxicological Laboratory investigations/confirmation of suspected toxin, it is very much essential to submit proper/correct supportive information along with labeled samples, in a leak proof, labeled container.

- d. While conducting the postmortem, it is necessary to collect the history from the Livestock owner, so as to rule out the possibilities of specific poison, which is expected from your end.
- e. So also information regarding, breed, age, sex, flock size, mortality rate, pregnant, milking, feeding/ grazing practices, consumed lush green grass/plants or the moulded feed/fodder (since how many days/months), kind of feed/fodder regularly given, or is there any change in feed and feeding practices, detailed symptoms, treatment undertaken.

The economic impact is great when there is loss of multiple animals in a herd due to poisoning. Every effort should thus be made to increase the availability of life-saving antidotal therapies for these animals. Stakeholders (materially concerned in its welfare) from both within and outside government are encouraged to participate in efforts to develop an acceptable way to resolve this important issue.

Antidote - An antidote is a substance which can counteract the activity or effect of a poison. Antidotes may be classified according to their mechanism of action: -

1. Chemical -

An antidote interacts specifically with a toxicant, or neutralizes the toxin. For example, metal chelators combine with metals to form complexes that can then be eliminated by the kidneys.

2. Pharmacologic -

Antidotes may neutralize or antagonize the effects of a toxicant. Antidote may act by preventing the formation of toxic metabolites by facilitation of more rapid or complete elimination of a toxicant (e.g. molybdenum for copper toxicity), by blocking receptors responsible for the toxic effect (e.g. Atropine-organophosphate toxicity), In addition, specific antidotes may act directly on the toxin (e.g. Antitoxins)

Many a times the cause of poisoning can not be traced out. It is not practical for the field Veterinarians to wait for the Lab diagnosis & guidelines of the treatment of acute cases of poisoning. Therefore, it is necessary to be familiar with the basic principles & the general line of treatment to be adopted in treating the poisoning cases.

a) To prevent further exposure to the Poison/Toxin: -

First step is to remove the affected animal from the place of exposure and change in the source of feed, fodder and water. This can limit the further exposure to the poison/toxin.

b) Removal of unabsorbed poison: -

When it is known that the toxicant is consumed by the animal, wash thoroughly the body surface of the animal with cold water is indicated. *Use of soap, alcohol & oily solutions should be avoided as they may promote the absorption.* Sometimes the washing may be difficult in some of the poisonings. The contaminated wool, hair coat may be removed by shearing/shaving.

c) Removal of unabsorbed poison from the gut: -

When it is known and confirmed that some toxicant is consumed by the animal, the procedure differs from species to species. In dogs, Cats and Swine - Vomition may be induced to empty the stomach by administration of Emetics, Apomorphin hydrochloride (0.09mg/kg BW S/c) is generally a drug of choice.

Remedies -

- 1) **Household** - These remedies include mustard seeds and water or a strong solⁿ of salt or large crystals of Washing soda.
- 2) **Vomition** - should not be induced when there is a reason to believe that the stomach wall may be weakened or when there is possibility that the retching would rupture blood vessels. There may be a danger of inhalation pneumonia, if the patient does not have full control on the throat.
- 3) **Gastric lavage** - may also be helpful in the small animals but there must be cognizance of the risk of precipitating convulsions in poisons such as Strychnine or chlorinated hydrocarbon insecticides.
- 4) **Ruminants** - Only effective way to empty the rumen is by ruminotomy. But, due consideration must be given to the ability of animal to withstand the stress of such an operational procedure. After ruminal contents have been removed, they should be replaced with fresh food & appropriate volume of fluid, preferably physiological saline. Inoculation of the rumen slurry from healthy animal is desirable.
- 5) **Purgation** - may be of some value, particularly, when slow acting poisons are known to be involved. While giving the purgation, make sure that constipation does not interfere with the process of elimination of poison.
- 6) No attempt should be made to use severe cathartics (Purgative). The safest cathartic is Sodium Sulphate. Magnesium Sulphate & Castor oil may not be used. Because, Magnesium depresses the modularly center, it is not removed by kidneys and oil facilitates the absorption of poison.

- 7) **Inactivation of the unabsorbed poison** – Any poison remaining in the stomach/rumen efforts should be made to neutralize as far as possible by administration of suitable antidote. If nature or source of toxin/poison is known, a specific antidote can be given to the ailing animal suspected for poisoning. If nature or source of toxin/poison is unknown, symptomatic treatment with supportive treatment can prove useful to save the animal. Fluid therapy, antihistaminic, liver tonics etc. will also help in elimination of toxin via urine.

1. Overall & general treatment undertaken for Aflatoxicosis –

To minimize the effects of aflatoxin in affected animals & birds, following measures can be undertaken –

1. Immediate change of feed will reduce the mortality & the effects
2. Supplement of Vit. A, proteins, fats, & trace minerals will reduce the mortality rate.
3. Mortality can be minimized by giving Sulphur containing amino acids like, methionine, cystine
4. Broad spectrum antibiotics to prevent secondary bacterial infections.

Control measures of Aflatoxicosis:-

The incidences of Aflatoxicosis are increasing, because of increased activity in Cattle and poultry farming. While Aflatoxins are not cumulative in the animal system, repeated exposure to Aflatoxins will produce a cumulative effect on the economic performance of the farmer.

There is no specific cure for the already produced effect of Aflatoxin poisoning. Immediate change from incriminated feed to uncontaminated feed will minimize the mortalities. To increase the resistance and minimize the hepatotoxic effects of Aflatoxins, use of sulphur containing amino acids, such as methionine, cystine and lecithine supplemented by vitamin - A are being practiced to reduce the economic losses. This leaves alternative approaches, such as preventing contamination of ingredients used for feeds by toxigenic fungi and the result and accumulation of Aflatoxin in them.

Storage practices of the feed / Ingredients need to be modified to prevent fungal growth. There are several anti-fungal agents that can reduce fungal infection. Use of ammonia and propionic acid (1%) in stored corn and groundnut has been tried successfully.

Other fumigants are available that can interfere with the production of toxins. But these methods do not reduce or remove the toxin that is already present in the feed or feed ingredients. Decontamination of Aflatoxin by using physical, chemical and biological methods can also be practiced. Moisture content of the feed/ingredients should not be more than 11-14 %.

Physical: - Heat, Sun light and ultraviolet light.

Chemical: - Inactivation can be carried out by using aldehyde. Ammonia in varying concentrations appears to be more promising & detoxification depends on the duration, quantity & the temperature at which it is used.

Biological: - Attempts have been made to use microbial agents to destroy/ transform Aflatoxin in the feed.

2. Gossypol poisoning Treatment –

1. No specific treatment is known
2. Remove the animal from suspected diet
3. Activated charcoal and purgative

3. HCN Poisoning Treatment –

1. Sodium Nitrite 20 gm and Sodium thiosulphate 30 gm mixed in 500 ml distilled water, shall be given Intra Venous very slowly at the rate of 20 ml per 50 kg body weight.
2. Sodium thiosulphate 20% solution shall be given I / V very slowly @ of 10 ml per 50 kg body weight.
3. Sodium thiosulphate 30 to 60 gm orally, at hourly interval for 4 – 5 times
4. Methyl blue- 1% solution shall be given I / v very slowly at the rate of 1-2mg/kg body weight
5. Supportive treatment – Supplementation of vitamins & minerals. Calcium borogluconate/dextrose saline with antihistamine injection shall be given I / v very slowly

Control Measures:-

1. Hungry cattle, sheep & goats should not be allowed to graze in the harvested Jowar crop field, immature sprouts of Jowar contains HCN.
2. Sheep & goats shall not be fed only & fully on Hiver seeds, which also contain HCN.

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3. Sprouts of Jowar (*Dura*) shall grow to the flowering stage and then the animals are allowed to graze/consume the green Jowar fodder.
4. Linseed meal may be fed in small quantity without soaking and gruel the containing linseed should be thoroughly boiled to drive off free HCN
5. While animals are at work in the immature sprouts of Jowar field, mouth gouge shall be applied.
6. A veterinarian shall be called immediately, if undoubtedly, the animals found grazing in the Jowar field of affective (40-45 days old) age.

4. Jute Poisoning -

1. Activated Charcoal: -

Attempts are in progress to ascertain the specific antidote. However, in absence of knowledge about the specific antidote "Powdered Charcoal" or "Activated Charcoal" is found to be the most effective universal antidote. The dosages are as follows -

Sr No	Species	Dose
1	2	3
1	Cattle & Horses	200 to 225gm
2	Buffalo	250 to 300gm
3	Sheep & Goat	25 to 30gm
4	Calves	25 to 30gm
5	Pigs & Dog	20 to 25gm

The above mentioned antidote shall be given 2 to 3 times a day, thoroughly mixed with water. If there are no symptoms of gastroenteritis, administration of "Activated Charcoal" may be followed by a suitable smoothening, non irritant purgative.

2. Methylene Blue: -

As a general antidote, Methylene Blue in the doses of 5.00mg per kg BW dissolved in sufficient quantity of distilled water and injected i/v can be used.

3. Sodium Hyposulphate (Thiosulphate): -

May also be tried as a general antidote, in combination with Sodium Nitrate in doses i/v as follows -
(Hypo - may be given orally in suitable doses as in case of HCN Poisoning)

Cattle & Buffalo - Sodium Hyposulphate (Thiosulphate) - 15gm
- Sodium Nitrate - 3gm
- Distilled water - 200ml

Sheep, Goat & Pig - 1/5th of dose of large animal i.e. 40ml

Symptomatic Treatment: -

Along with Universal antidote, Symptomatic Treatment may also be undertaken as follows -

- i) To check dehydration - Glucose saline
- ii) Soothing drugs - to treat gastroenteritis
- iii) Even though, Nervous depression is not noticed & if animal is seen to be depressed - Stimulants can be given
- iv) If there are convulsions - Use Tranquilisers or sedatives

Supportive Therapy: -

- i) In Convulsions - 20% Dextrose saline
- ii) Liver corrective inj e.g. Belamyl, Livogen etc.
- iii) Antihistaminic inj wherever felt necessary

Important Note: - Immediately remove the animals from suspected poison

5. Nitrate/Nitrite poisoning Treatment -

Treatment shall be undertaken considering the history and symptoms exhibited by the animal.

- 1) **Anoxic syndrome-** methylene blue - It reduces the methaemoglobin. But at a critical time, failures may be attributed, resulting into formation of additional methaemoglobin.
Cattle - 1 gm (0.25gm in sheep) in 25 ml sterile water for 113 kg body weight OR 4% solution of methylene blue shall be given I/V or S/c. In severe cases the treatment may be repeated, depending on the stage of recovery.

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- 2) Supplementation of Vit -C / ascorbic acid & its salts as reducing agents. It reduces the methaemoglobin. 5 to 20 mg/kg i/v advisable.
- 3) *In acute cases* - give the treatment with all possible speed.
- 4) *To overcome the deficiency of Vit A* - Lucerne is recommended or in feed doses of Vit A are to be administered.
- 5) *Supplementation of Vit E* - Nitrate/Nitrites increases the dietary requirement of Vit E, like Vit A. High concentration ration allows ruminal microbial activity to destroy more Vit E and therefore feed lot cattle are at risk. Feed lot cattle & truck load cattle benefit from Vit E supplement but animals on pasture generally do not unless there are excess Nitrate/Nitrites. Vit E is anti-oxidant & important for immunocompetence. Vit E is fat soluble, stored in the body but not long as Vit A.
- 6) Large doses of mineral oil may serve to slow down the absorption of nitrite and to aid in its elimination from the gut

Control: -

1. Ruminants likely to expose to nitrates should be given adequate carbohydrates in the diet.
2. Hungry / traveling animals shall not be allowed to graze in the unknown crop fields or sideway plants on the street.
3. Before feeding silage suspected of dangerous levels of nitrate should be allowed to sun dry/.
4. Animals shall not be allowed to graze in crop fields. If the animals are found to be mistakenly / unknowingly seen grazing in such crop field, shall be removed at the first sign of poisoning.

6. Insecticide/Pesticide Poisoning Treatment -

Atropine is highly effective physiological antidote in Insecticide/Pesticide poisoning.

Doses: -

1. **Cattle** - 15mg in 10ml water i/v followed by 45mg in water s/c (0.25 to 0.5 mg/kg BW).
A higher dose may be given as per the severity of poisoning i.e. 1mg/kg BW,
Half dose - i/v and remaining half by s/c.
2. **Sheep & Goat** - 1.1 to 2.2 mg/kg BW, one third dose-i/v and two third by s/c.
3. **Horses** - total average dose - 65mg
4. **Dogs** - 0.4 - 4.0mg depending upon size & weight, by s/c or 0.22mg/kg BW
(Total average dose-2mg)
 1. The doses must be adjusted to the individual case and weight & size of the animal. While treating cats & dogs heart response should be checked by stethoscope.
 2. Efforts should be made to reduce down the level of poisoning to the vanishing point without recurrence/appearance of clinical signs.
 3. On the other hand, a reduction to half may lead to severe poisoning and even death.
 4. The key factor appears to be the rate at which the reduction in activity is brought about.
 5. A rapid reduction gives rise to acute signs but a gradual decrease may have minimal effects.

7. Oxalate Poisoning Treatment -

1. Inj. Calcium borogluconate-I / v or S / c,
2. Repetitions of treatment depend upon symptoms/stage of recovery.
3. Ancillary - Provision of ample of fluids to decrease precipitation of oxalate crystals in urinary tract.

Control:-

1. Hungry animals should not be allowed to feed on large quantity of toxic plants.
2. Water intake affects the oxalate intake i.e. give water in sufficient amount
3. Alternative feed/fodder should be given, if the animal known to be ingesting oxalate containing feed/fodder or greens or if the symptoms of oxalate poisoning seen.
4. Prophylaxis - Give dicalcium phosphate, other calcium salts and bone meal are ineffective
5. Adequate salt should be made available.

8. Urea Poisoning Treatment -

1. **Large ruminants** -
Immediate treatment is required, with Vinegar 5% solution (Acetic acid), as a antidote up to 4 lit.
In divided doses, as the clinical signs tend to recur 30 min after treatment.
2. **Small ruminants** -
Immediate treatment is required, with Vinegar 5% solution (Acetic acid), as a antidote up to 0.5 to 1 lit. In divided doses, as the clinical signs tend to recur 30 min after treatment.
3. **Effective treatment**- emptying the rumen either via large bore stomach tube or by rumenotomy.

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- 2) Supplementation of Vit -C / ascorbic acid & its salts as reducing agents. It reduces the methaemoglobin. 5 to 20 mg/kg i/v advisable.
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2. **Small ruminants** -
Immediate treatment is required, with Vinegar 5% solution (Acetic acid), as a antidote up to 0.5 to 1 lit. In divided doses, as the clinical signs tend to recur 30 min after treatment.
3. **Effective treatment**- emptying the rumen either via large bore stomach tube or by rumenotomy.

9. Copper poisoning Treatment -

Identify dietary sources of copper, and if possible remove them. Remove copper from all minerals. Individual treatment with ammonium molybdate & sodium thiosulphate can be effective but may not be economic. Other supportive therapy such as fluid therapy and antibiotics is of limited value

Prevention of Copper poisoning -

It has not yet been clearly established that, what could be the cause of the increased rate of copper poisoning in cattle. However, the risk of copper poisoning can be reduced by -

- a) Ensuring you know what the copper intake of your cows. Pay particular attention to the copper content of your mineral and your forage
- b) Using chelates with care. So-called 'organic copper' may be better absorbed than inorganic copper, but this increase the risk of copper toxicity, particularly if same amount of copper is fed.
- c) Do not supplement with copper unless you have clear evidence of copper deficiency (or molybdenum toxicity)

10. A universal antidote can be prepared as follows -

- i) Activated charcoal - 2 parts
- ii) Magnesium oxide - 2 parts
- iii) Tannic acid - 1 part
- iv) Kaolin - 1 part

Dosage -

1. Cattle & Horse - 250 gm 2 to 3 times a day
2. Calves - 30 gm 2 to 3 times a day
3. Sheep/Goat/Pig - 15 gm 2 to 3 times a day
4. Dog - 30 gm mixed in 300 ml water, administered within 24 hours.

In cattle, sheep & goats, this treatment should be followed with purgative - Magnesium Sulphate - 450 gm and 120 gm in sheep & goats.

Antidote Toxicity Withdrawal Times -

Measures must be taken to ensure that substantially extended withdrawal periods are established for the treated animal prior to marketing of milk, meat, eggs, and other edible products. Farmers must be given the scientific information, regarding the extended withdrawal times, so that he is expected to take appropriate measures to prevent the animal and its food products from entering the human food supply.

1. Atropine sulfate Organophosphate toxicity Single dose (0.03-0.06 mg/kg) IM, SQ, IV in cattle sheep and swine: 3 day milk and 14 day meat. Multiple doses up to 0.2 mg/kg: 6 day milk and 28 day meat.
2. Epinephrine Anaphylaxis Zero day for cattle, sheep and swine.
3. Vitamin K₁ Anticoagulant rhodenticide, sweet-clover toxicity Zero day WDT at 0.5-2.5 mg/kg IV, IM, SQ for cattle, sheep, goats and swine.
4. Dimercaprol Arsenic, lead and mercury toxicity Minimum 5 day milk and 5 day meat
5. Copper glycinate Molybdenum toxicity 30 day meat for cattle; not for use in dairy cattle
6. Calcium disodium EDTA Lead toxicity 2 day milk and 2 day meat for food animals
7. Activated charcoal Adsorbent Zero day milk and meat for food animals
8. Sodium nitrite, Sodium thiosulphate, Sodium sulfate - Cyanide toxicity 48 hour milk and 24 hour meat for food animals
9. Molybdic acid salts (for copper toxicity) 30-day meat
10. Methylene blue (for nitrate, nitrite and chlorate toxicity) 180-day meat.

Molybdic acid salts & methylene blue are not recommended for milking animals. Recently, methylene blue has been recognized to have carcinogenic activity as reasonably anticipated to be a human carcinogen.

Collection of suitable material/tissues for Lab Examination -

See the **Appendix-1** attached here with for required material/tissue/ feed/ fodder/green/dry as per the poisoning suspected and shall be sent to the Laboratory for further investigations along with the history/ supportive information/predisposing factors responsible to cause the poisoning/case of Toxicity.

Actions & Uses of medicaments commonly used in the poisoning -

A) Action of inj Atropine sulphate & precautions to be taken while treating the animal

Atropine is highly effective physiological antidote in Organophosphorus poisoning.

Atropine sulfate as an injectable should not be in excess of 15 mg per dosage unit for cattle, goats, sheep, horses and pigs and not in excess of 0.6 mg per dosage unit for dogs and cats.

Action of Atropine and precautions to be taken while treating the animal: -

1. Although atropine is highly effective as an antidote, it is important to understand that it has no effect on the basic biochemical lesions. It acts only to block the toxic effects of OP compounds. All animals with atropine can not be saved.
2. If poisoning occurs due to slow acting compounds - continued administration of atropine may prove as exhaustive to the animal as the poisoning.
3. In this poisoning, the cholinesterase enzyme neutralization or inhibition takes place. Such cases do have increased tolerance for atropine, therefore to obtain proper response, normal dosing of atropine is to be increased considerably. In addition some animals may have ability to mobilize atropine. In such animals, atropine is to be administered more frequently than may have expected.
4. **Clinician's objective** - to administer only enough atropine to achieve atropinisation and consequently control the symptoms. He should take care that the poisoning due to atropine shall not take place.
5. **Mildly poisoned animals** - prophylactic dose of atropine required less than which is required in per acute cases.
6. When degree of poisoning is slight - i/m or s/c inj is sufficient
7. **In Per acute cases -**
 - a. 1/4th dose of anticipated total dose should be given i/v, to bring symptoms under control without delay.
 - b. I/v Atropine must be given slowly. The results of i/v inj are instantaneous, reaching about 10 min after administration. Within this 10 min administration of atropine by other routes may be accomplished.
8. Once the signs of poisoning are controlled, level of atropinisation may need to be maintained for several hours, even for a day or more. This will depend according to the type of OP compound & dose ingested. The requirement of atropinisation is to be declined, if the signs reappear or no signs appear.
9. **Contraindicated drugs**-Morphine, Succinylcholine, Theophylline & Tranquilisers of phenothiazine derivatives (Anthelminthics) e.g. Chlorpromazine & promazine Hydrochloride, Acepromazine

B) Ammonium Molybdate -

Orally, ammonium molybdate and sodium sulfate together provide an inexpensive alternative to the chelator D-Penicillamine. This combination not only prevents further copper intake, but also supports its elimination from the hepatocellular storage compartments. Serum copper levels have been determined to monitor the mobilization of copper following this antidote therapy.

C) Acetic acid -

The acid will reduce the high pH in the rumen, which was caused by the excess ammonia. Also, iced water administered orally is used to chill the rumen, reducing poisonous reactive activity.

Microbes living in the goat's rumen require feeding to produce enzyme byproducts. Thus, roughage given to goats provides food for the microbes. Enzymes break down proteins consumed by the animals & create ammonia. The ammonia is used by bacteria to create more protein, but excessive amounts are absorbed in the bloodstream and flow into the liver. If the liver is overloaded with ammonia, the rumen's functional ability becomes impaired and the animal develops ammonia toxicity.

Urea poisoning is treated by acidification of the rumen and cold water to slow the breakdown of urea to ammonia and limit the ammonia absorption. Nitrate poisoning is treated by conversion of the methaemoglobin back to hemoglobin by use of methylene blue. In animals that do not die, full recovery from either poisoning is generally evident by 14 hours post intoxication.

- D) **Boric acid** - It is used to repair wounds, eye application, long term acting insecticide & fungicide
- E) **Calamine** - This medication is used for local application to relieve pain, itching and discomfort from minor skin irritations. It also helps to dry the oozing and weeping caused by irritation.
- F) **Calcium borogluconate** - Dosage - 50 to 100 ml in sheep, 500 ml in cows - IV or S/C

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Usually, it is not enough to save animals because imbalance of other ions has occurred. Na⁺ & K⁺ liberated when Ca⁺⁺ precipitated may get alkalosis. Also oxalates have interfered with enzymes activated by calcium and magnesium, thus interfering with energy metabolism. Sheep don't respond to calcium, but still try. Salt hunger may predispose, so supply ample salt in diet and ample water for kidney function.

- G) **Calcium carbonate** - insoluble salt, CaCO₃, occurring naturally in shells, limestone & chalk and also used in more purified forms; used as an antacid and calcium replenisher and in the treatment of osteoporosis.
- H) **Calcium hydroxide** a salt, Ca (OH) 2, used in solution as a topical astringent.
- I) **Calcium Oxide** - Corrosively alkaline & caustic earth, CaO; is having various industrial uses & also a pharmaceutical necessity.
- J) **Calcium edetate (Ca-EDTA)** - Calcium ethylenediamine tetra-acetic acid; the disodium and dipotassium salts are commonly used as anticoagulants in the preservation of blood samples for hematology. A chelating agent used parenterally in the treatment of lead poisoning.
- K) **Fluids to flush kidneys** - dextrose solution IV Ample water for kidney function.
- L) **Glycerine** - It is used for external use for wound healing, as a mild antiseptic, very useful in large animal wound dressing, particularly if wounds are dirty & pocketing.

M) Liquor Ammonia -

The injection of ten drops of strong liquor ammonia, diluted with three parts of water, into the veins, has been recommended in cases of snake-bite. It may be useful possibly in bites of less poisonous snakes, but is of no utility in bites by the cobra or daboia. More especially where the stomach is relaxed and distended with gas. It also stimulates the intestines, and aids the expulsion of gas from them. It is therefore very useful in the flatulence and colic of children.

N) Magnesium Sulphate -

A major saline purgative in cattle (ruminants) but less useful in other species, small amounts are laxative, very suitable for the precipitation of lead in the intestine, excess sulphate produces purgation, precipitated lead sulphate is expelled out. Inj of Magnesium Sulphate has anaesthetic effect & in excess, euthanasia action.

O) Magnesium Oxide -

- It is prescribed for internal use as an alkali to counteract increased gastric acidity and in cases of acid poisoning it has a mild laxative effect.
- It Increases osmotic gradient in small intestine, which draws water into intestines and causes distention. These effects stimulate peristalsis and bowel evacuation.
- Magnesium also plays a role in the removal of toxic compounds such as ammonia, from the body, and helps regulate the heartbeat.
- Treating arsenic poisoning in a animal, comprises of -
 - a) Adding magnesium hydroxide and a small amount of a divalent or trivalent metal cation to the animal;
 - b) Adsorbing the arsenic poison to the doped magnesium hydroxide; and then -
 - c) Separating and removing the doped magnesium hydroxide with adsorbed arsenic from the animal, thereby reducing the concentration of arsenic in the human or animal to below an acceptable level.

P) Potassium permanganate (KMnO₄)-

Strong oxidizing properties, antiseptic, wound and mouth wash lotion, deodorant.

- a. **Alkaloid poisons** - Two actions which are made use of, though the advisability of such usages is debated, are that of antidotal effects against most alkaloid poisons.
Dose - 0.2gm/lit
- b. **Snake bite** - On the site of snake bite, make an incision, rub crystals of KMnO₄; the toxin is destructed by the action of the oxidation property of KMnO₄. KMnO₄ must be administered immediately within 5 min of poisoning.

Toxi Circular

Q) Sodium Sulphate - Saline purgative & diuretic

R) Sodium Nitrite & Sodium Thiosulphate -

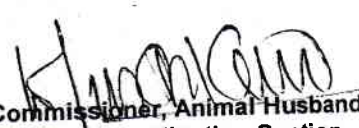
Dilatation of all peripheral blood vessels & coronary vessels, vasodilatation causes a drop in blood pressure & increase in heart rate, diaphoretic (Sweat producing) & antipyretic. Thiosulphate detoxifies the cyanide to harmless thiocyanate. Nitrites convert Hb into methHb which combine with cyanide to form cyanmeth Hb; it is a synergistic activity between these two antidotes.

S) Tannic acid -

Tannic acid has anti-bacterial, anti-enzymatic and astringent properties. Tannic acid has constricting action upon mucous tissues such as tongue and inside of mouth. The ingestion of tannic acid caused constipation and can be used to treat diarrhoea (in the absence of fever or inflammation). The anti-oxidant and anti-mutagenic properties of tannic acid are beneficial.

Internally - Tannic acid is styptic in intestinal hemorrhage and is valuable in treatment of diarrhea. It forms tannates when given as an antidote for poisoning by alkaloids and tartar emetic. Since these tannates are more or less soluble, however, some drug should be given as a purgative.

Encl - Appendix - 1, 2 and 3


Joint Commissioner, Animal Husbandry,
Disease Investigation Section,
Aundh, Pune - 411007
Maharashtra State

Copy for information & necessary action to -

1. Copy submitted for information to, Commissioner, Animal Husbandry, Maharashtra State, Pune -1
2. Regional Joint Commissioner, Animal Husbandry, **Mumbai / Nashik / Pune / Aurangabad / Latur / Amrawati and Nagpur** /- It is hereby requested to send copies of this circular to District Dy. Commissioner, Animal Husbandry, and District Animal Husbandry Officers in the region, so as to enable them to send the copies of the same to the Vety Institutes under their control.
3. Maharashtra Livestock Development Board, Akola
/- It is hereby requested to send copies of this circular to I/c Officer Bull Mother Farms (9), Bull Rearing Centers (3), F S laboratories (3) under the control of MLDB.
4. Punyashlok Ahilyadevi Sheli va Mendhi Vikas Mandal, Gokhalenagar, Pune-16
/- It is hereby requested to send copies of this circular to I/c Officers of Farms (9), under control.

Appendix - 1

General Guidelines for collection, preservation and dispatch of material/samples to be sent to the laboratory for examination, in various poisoning / toxicities

Sr No	Cases with history of ...	Order of preference-Material to be sent	Preservative to be used	Quantity of sample required
1	Hydrocyanic acid (HCN) poisoning/ Cyanide poisoning	Ruminal, Intestinal content, Vital organs - liver, lung, muscle piece fodder, greens, Blood	Common salt / saturated salt solution	Vital organs - 250-500 gm, Fodder/green - 250-500 gm, Feed/fodder/green/concentrates/grains - 250-500 gm, 15-20 ml
2	Nitrate / Nitrite	Suspected water, fresh urine, feed, concentrate, mineral mixture fodder, greens, plants, serum, blood & vital organs, Bones, teeth	No preservative Potassium oxalate + Sodium fluoride Immediately on ice within 1 - 2 hours OR on common salt/saturated salt solution	Vital organs - 250-500 gm, Feed/fodder/green/concentrates/grains - 250-500 gm, Water - 200 ml, Urine - 15-20 ml, Serum - 5-10 ml, Fodder/plants/green - 250-500 gm, Urine - 15-20 ml, Serum - 5-10 ml, Vital organs - 250-500 gm, Feed/fodder/green/concentrates/grains - 250-500 gm, Urine - 15-20 ml, Serum - 5-10 ml
3	Oxalate poisoning	Vital organs fodder, greens, plants serum, fresh urine	Common salt / saturated salt solution No preservative On ice	Feed/fodder/green/concentrates/grains - 250-500 gm, Urine - 15-20 ml, Serum - 5-10 ml
4	Alfatoxicosis	Feed/fodder/concentrates/grains	No preservative	Feed/fodder/concentrates/grains - 250-500 gm,
5	Zinc phosphide	Ruminal, Intestinal, stomach content, unopened crop, gizzard	In air tight container or sealed polythene bag	Ruminal, Intestinal, stomach content - 250-500 gm, and unopened crop, gizzard
6	Urea poisoning	Ruminal, Intestinal, stomach content Suspected chemical, feed, fresh urine, unclotted fresh blood (with preservative)	On ice No preservative	Ruminal, Intestinal, stomach content - 250-500 gm, Suspected chemical - 50 ml, Feed - 250-500 gm, Fresh urine - 15-20 ml, Unclotted fresh blood - 15-20 ml Feed/fodder/concentrates/grains/Suspected chemical-250- 500 gm
7	Insecticide /Pesticide Poisoning	Ruminal, Intestinal content, Vital organs - liver, lung, Kidney, Suspected poisonous material, water, feed, concentrate, mineral mixture, fodder, greens	Common salt / saturated salt solution No preservative	Vital organs - 250-500 gm, Feed/fodder/green/concentrates/grains - 250-500 gm,
8	Unknown poisoning	Ruminal /Intestinal/Stomach content, Vital organs - liver, kidney, lung, spleen, suspected poisonous material, Feed, fodder, greens, concentrate, grains	Common salt / saturated salt solution No preservative	Vital organs - 250-500 gm, Feed / fodder / plants / green / concentrates / grains - 250-500 gm, Water - 200 ml Feed/fodder/green/concentrates/grains/ suspected poisonous material - 250-500 gm,
9	Arsenic poisoning	Ruminal, Intestinal, stomach content	Common salt / saturated salt solution	Ruminal, Intestinal, stomach content - 250-500 gm
10	Common salt poisoning	Ruminal /Intestinal/Stomach content, Vital organs - liver, kidney, lung, spleen, part of external ear Feed,	On ice No preservative Potassium oxalate + Sodium fluoride	Ruminal /Intestinal/Stomach content - 250-500 gm Vital organs - 250-500 gm, Feed - 250-500 gm, part of external ear Feed/fodder/green/concentrates/grains - 250-500 gm, Blood 15-20 ml
11	Gaseous poisoning	Blood from ailing animal	No preservative Potassium oxalate + Sodium fluoride	Blood 15-20 ml

Appendix - 2

**List of Essential medicaments and equipments
required to prepare a "Toxicological emergency kit"**

A) I P Drugs -

Sr No	Name of Medicine	Name of Poisoning
1	2	3
1	Ca-EDTA Or Ca-Versenate, Sodium Citrate, Magnesium Sulphate,	Lead Poisoning
2	Potassium Permanganate (0.01 to 0.1 %), Copper Sulphate (0.1% Soln)	Phosphorus Poisoning
4	Ammonium Molybdate, Sodium Sulphate	Chronic Copper Poisoning
5	Sodium Nitrate, Sodium Thiosulphate	H C N Poisoning
6	Methylene Blue (2% Soln)	Nitrate/Nitrite Poisoning
7	Ammonium Sulphate	Fluorine Poisoning
8	Acetic Acid (Vinegar) 5% Soln	Urea Poisoning
9	Turpentine Oil	To relieve tympany
10	Common Salt	In small animals, salt solution induces Vomition. As a preservative for tissues to send to DIS.

B) Proprietary Medicines -

Sr No	Name of Medicine	Sr No	Name of Medicine
1	2	1	2
1	Inj Calcium Borogluconate	7	Inj Nikethamide (Coramine), Analeptic, Cardio, respiratory stimulant
2	Inj Glucose Saline, Ringer Lactate Soln	8	Inj Adrenaline
3	Inj Dexamethasone	9	Inj Vit E
4	Antihistaminic Inj	10	Inj Vit C
5	Inj Mannitol 20 % Soln in 5% Dextrose	11	Inj Vit A
6	Inj Atropine Sulphate	12	Inj Vit B Complex

C) Other Chemicals and Equipments -

Sr No	Name of supportive Chemicals and Equipments	Sr No	Name of supportive Chemicals and Equipments
1	2	1	2
1	Glycerin I P	8	Calamina preparata
2	Tourniquet	9	Kaolin
3	Activated charcoal	10	Dilute Liquor Ammonia
4	Liquid Paraffin	11	Boric Acid
5	Sodium bi-carbonate	12	Trocar & Cannula
6	Slides, Test Tubes, Empty wide mouth bottles (Plastic), Vials, Spirit Lamp, Spirit, Napkin, Torch	13	Enema pot, Soap, Full arm disposable gloves, I/v set, Syringes, needles, Scissors,
7	Magnesium oxide	14	Tannic acid

In above items, there are very few medicines, chemicals, injections which have expiry dates, as such periodic checkup of the kit would eliminate the use of date barred preparations. In case, if no emergency occurs, the contents of this kit can be used for routine treatment. If the contents of the kits are used for routine treatment, they shall be replaced.

Appendix - 3

In General Antidotal Treatment in Various Cases of Poisoning

Sr No	Kind of poisoning	Antidote Treatment
1	2	3
1	Acids	Magnesium oxide, Calcium oxide, Soap solution, Sodium carbonate
2	Alcohol	Stimulants
3	Alkalis	Vinegar, Citric acid, Fat oils, sedatives
4	Urea/Ammonia	Dilute Vinegar, Demulcents, Sulphides
5	Antimony	Purgative, Tannic acid, Magnesium oxide, BAL (Dimercaprol)
6	Arbus precatorius	Symptomatic
7	Arsenic	BAL (Dimercaprol), Sod Thiosulphate, Moist Ferric hydroxide
8	Arsenic Trioxide	Ferric hydroxide
9	Barbiturate	Migimide
10	Calcium	Magnesium
11	Caster beans/leaves	Symptomatic
12	Chlorinated hydrocarbon	Barbiturate, Chloral hydrate, Calcium borogluconate, Diazepam
13	Copper	Diethyl Dithio Carbonate, Ca Versenate, Penicillamine
14	Cyanide	Sodium Nitrite & Sodium Thiosulphate in combination, 4-dimethylaminophenol
15	Ergot	Symptomatic
16	Fluorine	5% Ca preparations, Aluminum sulphate/chlorate as preventive in feed to form insoluble compounds which are excreted through faeces
17	Jute	Methylene blue
18	Lantana Camara	Glucose saline, Antihistaminic, Liver tonics
19	Lead	Ca - EDTA, CNS depressant, Magnesium sulphate
20	Molybdenum	Copper sulphate
21	Nicotine	Orally - Tannic acid, Stimulants
22	Nitrate/Nitrite	Methylene blue
23	Insecticide/Pesticides	Atropine sulphate, PAM (Pralidoxime)
24	Oxalic acid	Orally - Lime water
25	Phenol	Gastric Lavage, Olive oil, egg albumin, skin wash with soap & water
26	Phosphorus	Gastric Lavage, Saline purgative, Weak soln of Copper sulphate
27	Pilocarpin	Atropine
28	Selenium	Sod arsenate
29	Silver	Sod Chloride
30	Snake bite	Polyvalent Antivenin, Anti Tetanus Serum, Antibiotics, Corticosteroids, antihistaminic
31	Sodium Chloride	Salt free water, Ca Gluconate
32	Strychnine	Orally Pot Permanganate, Barbiturate I/v
33	Toads	Corticosteroids, Sedatives, Sod Carbonate
34	Zinc	BAL (Dimercaprol), Orally - Sod & Potassium Carbonate in water
35	Zinc Phosphide	Symptomatic